



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
08-08-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

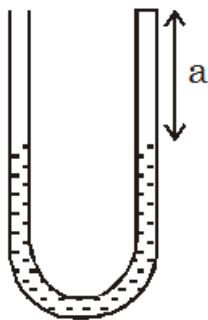
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A wave $y = 6 \text{ mm} \sin(100 \pi t - \pi x)$ is travelling in a string towards a boundary. Beyond the boundary, the mass per unit length of the string becomes four times whereas the tension remains the same. Assume that the boundary is at $x = 0$. The common tension in both the strings is 100 N. Choose the correct statement(s). (x is in m & t is in sec).

- (A) The equation of reflected wave is $y = -2 \text{ mm} \sin(100 \pi t + \pi x)$
- (B) The equation of transmitted wave is $y = 4 \text{ mm} \sin(100 \pi t - 2 \pi x)$
- (C) The average power of the incident wave is 1.8 W
- (D) The average power of reflected wave is 0.6 W

Space for Rough Work

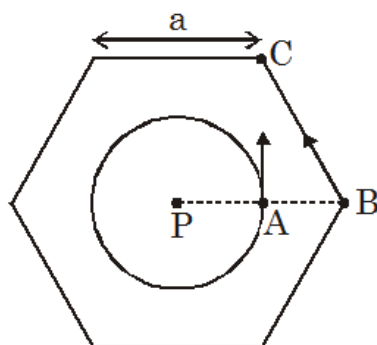
2. The equations : $y_1 = a \sin(\omega t - kx)$ and $y_2 = -a \sin(\omega t + kx)$ are for two waves, then the waves will :
- (A) Move in the same direction with equal amplitude
- (B) Move in opposite direction with the same speed and the same frequency
- (C) Produce beats by superposition
- (D) Produce stationary waves by superposition
3. A long U tube is filled with a liquid of density ' ρ ' such that length of the tube above liquid is ' a ' in both arm. One side of tube (right arm) is sealed and the tube is inverted (' P_0 ' atmospheric pressure). The liquid is disturbed so that it starts moving from right hand side to left hand side.



- (A) Liquid will spill out the left tube if $a < \frac{P_0}{4\rho g}$
- (B) Liquid will not spill out the left tube for any value of ' a '
- (C) Liquid surface in left arm will not move if $a = \frac{P_0}{2\rho g}$
- (D) Liquid surface in right arm will go up.

Space for Rough Work

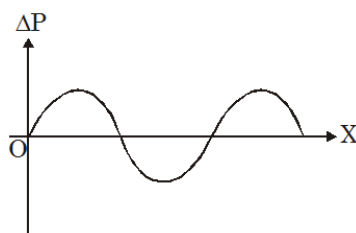
4. A boy is moving uniformly along a circular track in anticlockwise sense. A point sound source moves along a regular hexagonal path. The centres of hexagonal track and the circular track coincide as shown in the figure. The source and boy starts from point B and A respectively such that P, source & boy always lie on the same radial line. Neglect the time taken by sound to reach the boy.



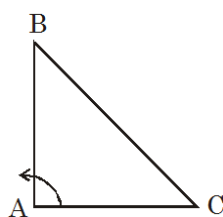
- (A) Just after B, the frequency heard by the boy is maximum.
 (B) Just after C, the frequency heard by the boy is minimum.
 (C) At the mid-point of BC, the boy hears the original frequency.
 (D) Just after B, the frequency heard by the boy is minimum.

Space for Rough Work

5. A sinusoidal longitudinal wave is travelling in positive x direction. Wave length of the wave is 0.5 m. At time $t = 0$, the change in pressure at various points on the x axis can be represented as shown in figure. Consider five particles of the medium A, B, C, D and E whose co-ordinates are 0.125 m, 0.1875 m, 0.250 m, 0.3757m and 0.50 m respectively.



- (A) A and B are moving in positive x direction at $t = 0$.
- (B) C and D are moving in positive x direction at t such that $\omega t = \frac{\pi}{2}$. Here ω represents the angular frequency.
- (C) The ratio of initial speed of particles at point A and C is 0.5
- (D) The amplitude of oscillation at E is half the amplitude of oscillation at C.
6. Consider a uniform isosceles right angled plate ABC. It's moment of inertia about A perpendicular to plane of the plate is I . It's moment of inertia about B perpendicular to the plane is I_1 . Its moment of inertia about line AB is I_2 and about line BC is I_3 .



- (A) $I_1 = 4I$
- (B) $I_2 = \frac{I}{2}$
- (C) $I_3 = \frac{I}{4}$
- (D) None of these

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. A tuning fork is placed near a vibrating stretched wire. If frequency of tuning fork is greater than frequency of stretched wire, then a boy standing near the two hears a beat frequency f .

List-I		List-II	
(P)	If tuning fork is loaded with wax	(1)	Beat frequency must increase
(Q)	If prongs of tuning fork are filed	(2)	Beat frequency must decrease
(R)	If tension in stretched wire is increased	(3)	Beat frequency may increase
(S)	If tension in stretched wire is decreased	(4)	Beat frequency may decrease

(A) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3, 4; S \rightarrow 1$

(B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3, 4; S \rightarrow 1$

(C) $P \rightarrow 3, 4; Q \rightarrow 1; R \rightarrow 3, 4; S \rightarrow 1$

(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3, 4; S \rightarrow 1$

8. A rigid body of mass M and radius R rolls without slipping on an inclined plane of inclination θ , under gravity. Match the type of body with magnitude of the force of friction.

List-I		List-II	
(P)	For ring	(1)	$\frac{Mg \sin \theta}{2.5}$
(Q)	For solid sphere	(2)	$\frac{Mg \sin \theta}{3}$
(R)	For solid cylinder	(3)	$\frac{Mg \sin \theta}{3.5}$
(S)	For hollow spherical shell	(4)	$\frac{Mg \sin \theta}{2}$

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 4$

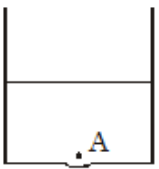
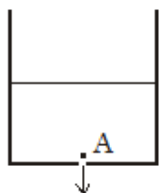
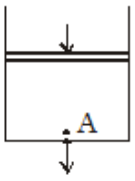
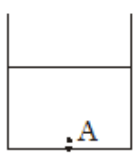
(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

Space for Rough Work

9. A container contains water filled upto a height h . There is a small hole at the bottom. Match the cases listed in list-I with the pressure at point A in list-II.

List-I		List-II	
(P)	Water does not leak out due to surface tension. Water is being added slowly. 	(1)	Pressure increases with time
(Q)	Water flows out. Neglect effect of surface tension 	(2)	Pressure decreases with time
(R)	Water is being pushed out with constant speed by using a piston. Neglect effect of surface tension. 	(3)	Pressure remains constant with time
(S)	The container is dropped to fall freely. 	(4)	Variation of pressure with time cannot be predicted.

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 3$

(B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 3$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

Space for Rough Work

10.

List-I		List-II	
(P)	A constant force acting along the line of SHM affects	(1)	the time period
(Q)	A particle falling and sticking on the block (spring + mass system) executing SHM on a smooth horizontal plane when the latter is at extreme position affects	(2)	the frequency
(R)	A particle falling and sticking on the block (spring + mass system) executing SHM on a smooth horizontal plane when the latter cross the mean position affects	(3)	the mean position
(S)	A particle connected to a spring whose other end is fixed on a smooth table. If the table is pushed with a uniform horizontal acceleration along the direction of the spring, the acceleration of the table affects	(4)	the amplitude

(A) $P \rightarrow 1,2,3,4; Q \rightarrow 1,2,3,4; R \rightarrow 1,2; S \rightarrow 3,4$

(B) $P \rightarrow 3,4; Q \rightarrow 1,2; R \rightarrow 1,2,4; S \rightarrow 3,4$

(C) $P \rightarrow 3,4; Q \rightarrow 1,2,3,4; R \rightarrow 1,2; S \rightarrow 3,4$

(D) $P \rightarrow 1,2,3,4; Q \rightarrow 1,2,3; R \rightarrow 1,2; S \rightarrow 3,4$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

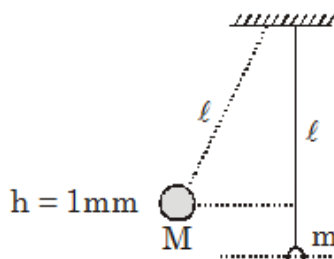
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. When a violinist pulls the bow across a string, the force with which the bow is pulled is fairly small, about 0.6 N. Suppose the bow travels across the A string, which vibrates at 440 Hz, at 0.5 m/s. A listener 30 m from the performer hears a sound of 60 dB intensity. With what efficiency (in percent) is the mechanical energy of bowing converted to sound energy? (Assume that the sound radiates uniformly in all directions.) $\pi = 3.14$
 2. The lid of a cuboid box is balanced vertically on its hinges. The slightest displacement from the vertical leads to its falling shut. If the lid is of uniform density and it is 30 cm wide, what will be its angular velocity (in rad/s) when it does fall shut, that is, when it is in a horizontal position?
 3. A uniform rod of mass 2 kg and length 3 m is kept on a smooth horizontal surface. One of its end is struck with an impulse of 4 Ns. What is the distance (in m) travelled by the centre of mass by the time the rod has rotated by 2 revolutions? $\pi = 3.14$
 4. A ball of radius 5 cm falls through air. The mass of the ball is 20 gm and we can safely neglect the buoyant force of the air. What is the rate (in kW) at which energy is being lost in steady state? Take the viscosity of air to be 2×10^{-5} kg/m sec. $\pi = \frac{22}{7}$
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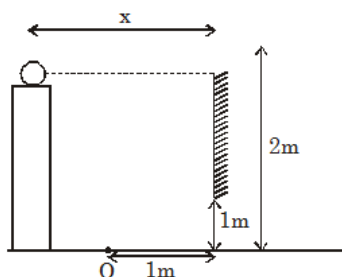
Space for Rough Work

5. A string of natural length 0.99 m is tied between two poles separated by 1.00 m. The string has a cross section area of 1 mm^2 and Young's modulus of $1.98 \times 10^{11} \text{ N/m}^2$. The string has a density of 8000 kg/m^3 . What is the frequency of first overtone (in Hz)?
6. Two pendulums of length ℓ are hanging next to each other, such that the balls at their ends are touching each other as shown in the figure. The mass of one of the balls, M , is much greater than that of the other, m . Take $\ell = 1 \text{ m}$.



The pendulum on the left is displaced a bit, such that the elevation of the ball of mass M is 1 mm , and then it is released. After the first elastic collision of the balls what maximum height (in mm) from its original position can the ball of mass m reach?

7. A stationary plane mirror has normal on the reflecting surface along the direction $2\hat{i} + 2\hat{j} + \hat{k}$. If an object in front of the mirror is moving with a velocity of $3\hat{i} + 4\hat{j} + 4\hat{k} \text{ m/s}$, what is the x component of the velocity (in m/s) of the image?
8. An observer stands in front of a plane mirror kept in a vertical plane. His eyes are at the same level as the edge of the plane mirror. At what maximum distance x (in m) can he stand in front of the mirror so that he can see the image of O ?



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A metal 'M' crystallizes in two crystal systems fcc and hcp. If d = density, z = no. of atoms per unit cell and a, b, c are edge lengths of the unit cell along the crystallographic axes then. Select the correct options :

(A) $\frac{d_{\text{fcc}}}{d_{\text{hcp}}} \neq 1$

(B) $\frac{a_{\text{fcc}}}{c_{\text{hcp}}} = \frac{\sqrt{3}}{2}$

(C) $\frac{d_{\text{fcc}}}{d_{\text{hcp}}} = 1$

(D) $\frac{z_{\text{fcc}}}{z_{\text{hcp}}} = \frac{2}{3}$

Space for Rough Work

2. A vessel initially contains two gases A and B in the mass ratio 2 : 1 respectively. If the above gas mixture is allowed to effuse through a hole then the molar composition of the mixture coming initially out of the hole is 1 : 4. Select correct option(s) :
- (A) The simplest whole number ratio of the moles of gases A and B present in the vessel in the beginning respectively is 1 : 2.
- (B) The simplest whole number ratio of the moles of gases A and B present in the vessel in the beginning respectively is 1 : 8
- (C) If the mass ratio of gases A and B required in the final concentrated sample is 1 : 32 then the number of successive steps of effusion to be carried out is 5
- (D) If the mass ratio of gases A and B required in the final concentrated sample is 1 : 32 then the number of successive steps of effusion to be carried out is 6.

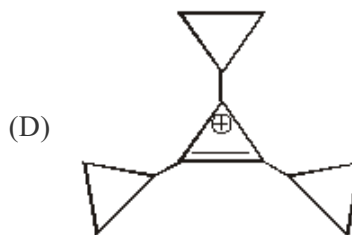
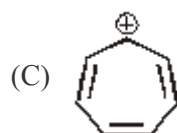
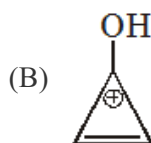
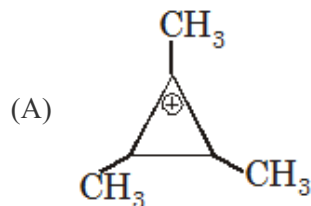
3. The compound  is an isomer of



Space for Rough Work

4.

Which of the following are more stable than



5. In which of the following pair(s), given property is not same for the both compounds.

(A) Diamond, Graphite (C-C bond length)

(B) α -sulphur, β -sulphur (Atomicity)

(C) α -sulphur, β -sulphur (Soluble in CS_2)

(D) White phosphorus, Red phosphorus (Chemiluminescence)

6. Select species in which $3d_{z^2}$ and $3d_{x^2 - y^2}$ orbitals involved in hybridisation of central atom?

(A) $\text{SH}_{6(g)}$

(B) $\text{XeF}_{6(g)}$

(C) $\text{PCl}_{5(s)}$

(D) $\text{ClF}_{5(g)}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

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Negative Marks : -1 In all other cases

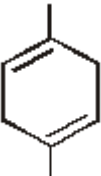
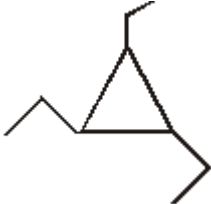
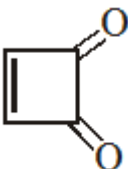
7. Boyle's temperature (T_B): T_B of $\text{CO}_2 = 350^\circ\text{C}$, T_B of $\text{CH}_4 = 60^\circ\text{C}$ and T_C of $\text{H}_2 = -240^\circ\text{C}$

List-I		List-II	
(I)	CO_2 gas in low pressure region & temperature 573 K.	(P)	$PV = nRT$
(II)	CO_2 gas in very high pressure region & temperature 623 K	(Q)	Gas is more compressible with respect to an ideal gas.
(III)	CH_4 gas at 333 K & in low pressure region.	(R)	Gas cannot be liquefied at any pressure at given temperature.
(IV)	H_2 gas in low pressure region & temperature 20 K.	(S)	Molar volume of gas is lesser than that of ideal gas at same pressure and temperature.
		(T)	Repulsive forces dominate

- (A) I $\rightarrow$ Q,R; II $\rightarrow$ R; III $\rightarrow$ P,R; IV $\rightarrow$ Q,S
- (B) I $\rightarrow$ Q,R,S,T; II $\rightarrow$ R,T; III $\rightarrow$ P,R; IV $\rightarrow$ Q,S
- (C) I $\rightarrow$ Q,R,S; II $\rightarrow$ R,T; III $\rightarrow$ P,R; IV $\rightarrow$ P,Q,S
- (D) I $\rightarrow$ Q,R,S; II $\rightarrow$ R,T; III $\rightarrow$ P,R; IV $\rightarrow$ Q,S

Space for Rough Work

8. Match the compounds in list-I with statements in list-II :

	List-I		List-II
(P)		(1)	Degree of unsaturation is even
(Q)		(2)	Ratio of $1^\circ\text{C} : 2^\circ\text{C} : 3^\circ\text{C}$ is 1 : 1 : 1
(R)		(3)	Five carbon-carbon π -bonds are present
(S)		(4)	Six primary hydrogen present

(A) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

(B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(C) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

(D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

Space for Rough Work

9.

List-I (Specie)		List-II (Characteristics of specie)	
(P)	IO_2F_2^-	(1)	All bond lengths are same
(Q)	SH_6	(2)	All bond angles are same
(R)	XeF_5^-	(3)	Molecule/ion do not exist
(S)	SO_4^{2-}	(4)	Non-planar
		(5)	$2p\pi-3d\pi$ type bond present

(A) $\text{P} \rightarrow 5,4; \text{Q} \rightarrow 3; \text{R} \rightarrow 1,2; \text{S} \rightarrow 1,2,4,5$

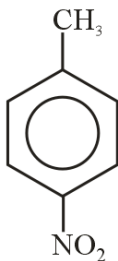
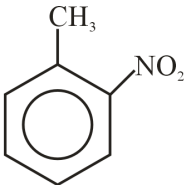
(B) $\text{P} \rightarrow 4; \text{Q} \rightarrow 3; \text{R} \rightarrow 1,2; \text{S} \rightarrow 1,2,4,5$

(C) $\text{P} \rightarrow 1,5,4; \text{Q} \rightarrow 1,2; \text{R} \rightarrow 1,2; \text{S} \rightarrow 1,2$

(D) $\text{P} \rightarrow 4; \text{Q} \rightarrow 1,2; \text{R} \rightarrow 1,2,4; \text{S} \rightarrow 1,2$

Space for Rough Work

10.

List-I (Comp.-1) (Comp.-2) (Comp.-3)				List-II (Order of mention property with group of compounds given)	
(P)	F ₂	Cl ₂	Br ₂ (Bond energy)	(1)	1 < 2 < 3
(Q)	C ₂ H ₄	C ₃ H ₄	C ₄ H ₄ (Dipole moment)	(2)	3 < 2 < 1
(R)	H ₂ O	NH ₃	HF (Boiling point)	(3)	1 = 2 = 3
(S)	CH₃  NO₂ CH₃  NO₂ CH₃  CH₃ (Dipole moment)			(4)	1 > 3 > 2
				(5)	2 > 3 > 1

(A) P → 2; Q → 3; R → 1; S → 4

(B) P → 2; Q → 1; R → 2; S → 2

(C) P → 5; Q → 3; R → 4; S → 2

(D) P → 5; Q → 3; R → 2; S → 1

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

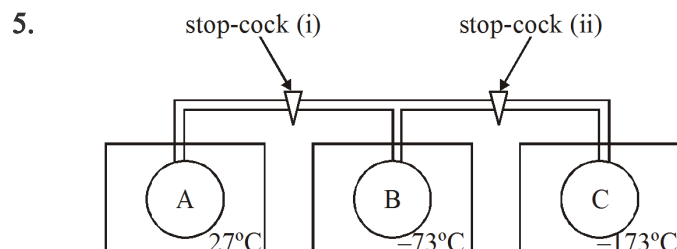
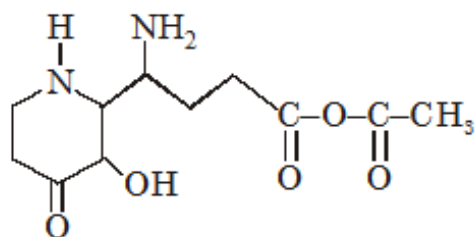
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A protein crystallizes as orthorhombic crystal system with 6 molecules of protein per unit cell. If density of protein is 1.5 g/ml. Calculate molar mass of the protein in kg/mole. The dimensions of orthorhombic unit cell is as follows :
 $a = 125\text{\AA}$, $b = 80\text{\AA}$ and $c = 30\text{\AA}$ (Take $N_A = 6 \times 10^{23}$)
2. A 1.25 g sample containing Fe_2O_3 , Fe_3O_4 and an inert impure substance is treated with excess of KI solution in presence of dilute H_2SO_4 . The entire iron is converted into Fe^{2+} along with the liberation of I_2 . The resulting solution is diluted to 100 ml. A 25 ml of the diluted solution requires 12 ml 0.5 M hypo solution to reduce the iodine present. A 50 ml of the diluted solution, after complete extraction of the iodine required 12.8 ml 0.25 M KMnO_4 solution in dilute H_2SO_4 medium for the oxidation of Fe^{2+} . Calculate mass % of Fe_2O_3 present in the original sample
3. The value of K_p for the reaction at 27°C
 $\text{Br}_2(\text{l}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{BrCl}(\text{g})$
 is '1' atm. At equilibrium in a closed container partial pressure of BrCl gas is 0.1 atm and at this temperature the vapour pressure of $\text{Br}_2(\text{l})$ is 0.13 atm. Then what will be minimum moles of $\text{Br}_2(\text{l})$ to be added to 1 mole of Cl_2 , initially, to get above equilibrium situation :

Space for Rough Work

4. How many different type of functional groups are present in following compound :



The above apparatus consists of three temperature jacketed 12.315 litre bulbs connected by stop cocks. Bulbs 'A' consist of $\text{NH}_3(\text{g})$, $\text{CO}_2(\text{g})$ & $\text{N}_2(\text{g})$ at 27°C and a total pressure of 60 atm. Bulb 'B' and 'C' are empty and maintained at temperature of -73°C and -173°C respectively. Initially stop-cocks are closed and volume of the connecting lines is zero.

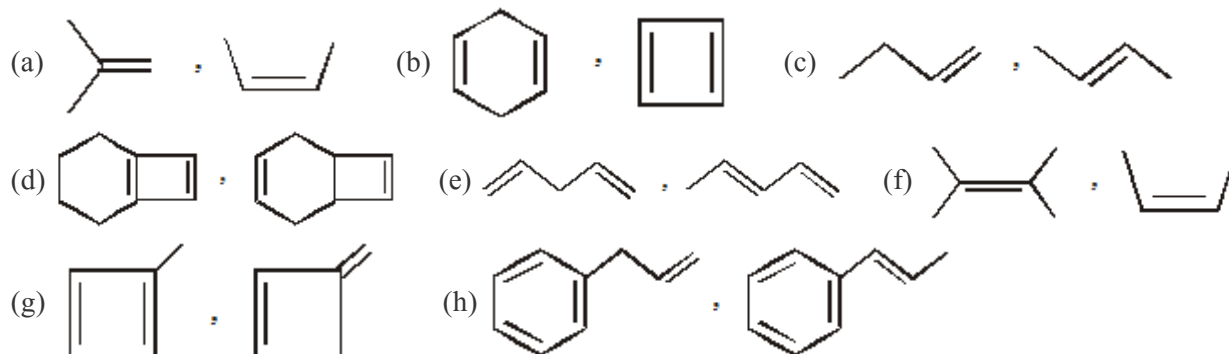
Given : ($R = 0.0821 \text{ L-atm/K-mol}$)

- $\text{NH}_3(\text{g})$ liquifies at -20°C , $\text{CO}_2(\text{g})$ liquifies at -80°C and $\text{N}_2(\text{g})$ liquifies at -180°C .
- Vapour pressure of $\text{NH}_3(\text{l})$ at $-73^\circ\text{C} = 1 \text{ atm}$
- Vapour pressure of $\text{NH}_3(\text{l})$ at -173°C is negligible
- Vapour pressure of $\text{CO}_2(\text{l})$ is negligible
- On opening stop-cock (i) only, the pressure in containers becomes 17 atm and, on opening both stop-cocks pressure decreases to 4 atm.

What is the value of $(n_{\text{NH}_3} + n_{\text{CO}_2} - n_{\text{N}_2})$ in the initial mixture ?

Space for Rough Work

6. Number of pairs in which 1st compound has more heat of hydrogenation than 2nd are :



7. In the parallel universe all quantum number values are same and all rules are same but value of azimuthal quantum number ranges from 0 to (n+1)

As per the given system, find atomic number of second inert gas in the parallel universe.

8. Calculate value of (x+y).

here, x is the number of valence shell electrons of Xe which are present in hybrid orbitals of XeO_3F_2 and y is the number of total lone pair in XeO_3F_2 molecule.

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. Consider the ellipse $\frac{x^2}{4} + y^2 = p$ and its given chord $x - y = 1$. For what values of p , the pair of tangents drawn from an external point to the ellipse meet the given chord at its extremities and the reflection of the external point about the given chord lie within the ellipse.
- (A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{4}{13}$ (D) $\frac{6}{13}$
2. Let two tangent to parabola at A and B on it are $x + 2y + 1 = 0$ and $2x - y + 7 = 0$ while normal drawn at A and B meet at C $(-2, -2)$ lying on the axis of parabola then which of the following is/are true
- (A) Equation of axis of parabola is $y + 3x + 8 = 0$ (B) Equation of axis is $y - 3x + 4 = 0$
- (C) Length of latus rectum $= \sqrt{10}$ (D) Length of latus rectum $= \sqrt{2}$
-

Space for Rough Work

3. Let $(4 + \cos\theta)^2 - 4\sin^4\theta = 25$ and number of possible values of θ for $\theta \in (0, 5\pi)$ is p . If $\theta_0, \alpha \in \left(0, \frac{\pi}{4}\right)$, $\cos^2\theta_0 - \frac{1}{p^2} = \sin^2\alpha + \frac{\sqrt{3}}{p^2}$ and $(\tan\theta_0 + 1)(\tan\alpha + 1) = \frac{p^2}{2}$, then which of the following statements is/are true ? ($\theta_0 > \alpha$)
- (A) $\theta = 2\alpha$ (B) $\tan 2\theta_0 \tan 2\alpha = 1$
- (C) $4 \sin\left(\frac{3\theta_0}{10}\right) = \sqrt{3 + \sqrt{5}} + \sqrt{5 - \sqrt{5}}$ (D) $\cos(2\theta_0 + \alpha) = \frac{\sqrt{3} - 1}{2\sqrt{2}}$
4. $y = x$ is tangent to an ellipse whose foci are $(1, 0)$ and $(3, 0)$ then
- (A) Length of major axis of ellipse is $= \sqrt{6}$
- (B) Length of major axis of ellipse is $= \sqrt{10}$
- (C) $\left(\frac{3}{4}, \frac{3}{4}\right)$ is the point of contact of this ellipse and this tangent
- (D) $\left(\frac{1}{2}, \frac{1}{2}\right)$ is the point of contact of this ellipse and this tangent

Space for Rough Work

5. A circle C_1 of radius 'b' touches the circle $C : x^2 + y^2 = a^2$ externally and has its centre on positive x-axis. Another circle C_2 of radius 'c' touches the circle C_1 externally and has its centre on positive x-axis. If $a < b < c$ and the three circles have a common tangent, then triplets (a, b, c) can be equal to
- (A) (2, 4, 8) (B) (2, 4, 6) (C) (3, 9, 27) (D) (7, 8, 9)
6. Let AD be a median drawn through vertex A to side BC of $\triangle ABC$ such that $\angle BAD = \theta$, $\angle DAC = 2\theta$ and $2\cos\theta = \cot B$ where $B \in \left(0, \frac{\pi}{2}\right)$. Which of the following is/are correct ?
- (A) $\angle A = \frac{\pi}{2}$ (B) Length of AD = $a \sin B$
- (C) Length of AD = $a \sin C$ (D) $\angle A = \frac{\pi}{3}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.

List-I		List-II	
(P)	If roots of $x^2 - kx + 36 = 0$ are integers then number of values of k equals	(1)	2
(Q)	Number of values of 'a' for which the common chord of the circles $x^2 + y^2 = 8$ and $(x - a)^2 + y^2 = 8$ subtends a right angle at the origin is	(2)	4
(R)	If $\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = \frac{k}{3}abc(a+b+c)^3$ then the value of k is	(3)	6
(S)	$\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$	(4)	8
		(5)	10

(A) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

(B) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

(D) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

Space for Rough Work

8.

List-I		List-II	
(P)	Number of solutions of equation $(1 - \tan \theta)(1 + \tan \theta) \sec^2 \theta + 2^{\tan^2 \theta} = 0$ where $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	(1)	2
(Q)	Find the value of the expression $2x^3 + 2x^2 - 7x + 72$ when $x = \frac{3 + 5\sqrt{-1}}{2}$	(2)	3
(R)	The number of solution of the equation $\cot x = \frac{\pi}{2} + x$ in $x \in \left[-\pi, \frac{3\pi}{2}\right]$ is	(3)	4
(S)	Number of integral values of λ , for which equation $4\cos x + 3 \sin x = 2\lambda + 1$ has a solution is	(4)	6
		(5)	8

(A) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(B) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

Space for Rough Work

9.

List-I		List-II	
(P)	P lies on the line $y = x$ and Q lies on $y = 2x$. The equation for the locus of the mid point of PQ, if $ PQ = 4$, is $25x^2 - \lambda xy + 13y^2 = 4$, then λ equals	(1)	5
(Q)	Number of integral values of λ if $(\lambda, \lambda + 1)$ is an interior points of $\triangle ABC$, where $A \equiv (0, 3)$, $B \equiv (-2, 0)$ and $C \equiv (6, 1)$.	(2)	2
(R)	If distance between the pair of parallel lines $x^2 + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ is $25\sqrt{2}$, then $ a $ is equal to	(3)	10
(S)	If radii of the smallest and largest circle passing through the point $(\sqrt{3}, \sqrt{2})$ and touching the circle $x^2 + y^2 - 2\sqrt{2}y - 2 = 0$ are r_1 and r_2 respectively, then the arithmetic mean of r_1 and r_2 is	(4)	36
		(5)	1

(A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 5$

(B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$

(C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 5$

(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 5$

Space for Rough Work

10.

List-I		List-II	
(P)	$\frac{\sqrt{31 + \sqrt{31 + \sqrt{31 + \dots}}}}{\sqrt{1 + \sqrt{1 + \sqrt{1 + \dots}}}} = a - \sqrt{b}$ where $a, b \in \mathbb{N}$, then the value of $(a + b)$ is	(1)	29
(Q)	If $n, m \in \mathbb{N}$ and $m = \frac{n^2 - n - 35}{n - 4}$, then the value of m is	(2)	16
(R)	Let $x = (\log_{1/3} 5)(\log_{125} 343)(\log_{49} 729)$ and $y = 25^{3 \log_{289} 11 \log_{28} \sqrt{17} \log_{1331} 784}$, then find the value of $(x^2 + y^2)$	(3)	11
(S)	Find the absolute value of product of all the integers which do not belong to the solution set of the inequality $\left \frac{3 x - 2}{ x - 1} \right \geq 2$.	(4)	1
		(5)	34

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(B) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 2; S \rightarrow 4$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 4$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A triangle ABC with vertices $B \equiv (5, 2)$ and $C \equiv (-1, -1)$ is inscribed on the curve $xy - x - y - 3 = 0$ and $H \equiv (\alpha, \alpha)$ is the orthocentre of $\triangle ABC$. Let D, E and F be the feet of perpendicular from A, B and C on BC, AC and AB respectively. If the inradius of $\triangle DEF$ is expressed in the form of $\frac{\sqrt{\lambda}}{5}$ then value of $\frac{\lambda}{5}$ equals
2. A straight line through the point $P(\alpha, 2)$ where $\alpha \neq 0$ meets the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D and meets the x & y axes at B and C respectively such that $PA \cdot PD = PB \cdot PC$ then find the minimum value of $\frac{|\alpha|}{5}$.
3. A variable line 'L' is drawn through (0, 0) to meet the lines $L_1 : y - x - 10 = 0$ and $L_2 : y - x - 20 = 0$ at points A and B respectively. A point P is taken on L such that $\frac{1}{OP^2} = \frac{1}{OA^2} + \frac{1}{OB^2}$. If the locus of P is $|y - x| = a\sqrt{b}$. Determine the least value of $\frac{(a + b)}{4}$ (where a, b $\in \mathbb{N}$)
4. If the co-ordinate of the vertex of the parabola whose parametric equation is $x = t^2 - t + 1$ and $y = t^2 + t + 1$, $t \in \mathbb{R}$ is (a, b), then (2a + 4b) equals

Space for Rough Work

5. If the value of $\left(\tan^2 \frac{\pi}{7} + \tan^2 \frac{2\pi}{7} + \tan^2 \frac{3\pi}{7}\right) \left(\cot^2 \frac{\pi}{7} + \cot^2 \frac{2\pi}{7} + \cot^2 \frac{3\pi}{7}\right)$ is λ then determine $\frac{\lambda}{42}$
6. Find the number of ordered pair (p, q) where $p, q \in \mathbb{N}$ if the equation $(x^2 - px + q)(x^2 - qx + p) = 0$ has all roots as natural numbers.
7. Let $f(x)$ be a polynomial of degree 7 with leading coefficient unity and $f(1) = 1, f(2) = -5, f(3) = -9, f(4) = -11, f(5) = -11, f(6) = -9$ and $f(7) = -5$, value of $\frac{|f(8)|}{1000}$ equals
8. Let focal chord intersect hyperbola at $(1, 5)$ and $(6, 0)$ and the equation of tangent at $(1, 5)$ is $3x + y = 8$ and equation of tangent at $(6, 0)$ is $x + 2y = 6$. If the length of latus rectum is expressed in the form of $\frac{a\sqrt{2}}{b}$ (where a and b are co-prime natural numbers) then find $\frac{(a - 2b)}{5}$.

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work